a

Component	Sources	Bench-measurable	Mitigation (evidence)	Residual risk (this report)	Kind (this report)	Cross-domain (a9)
Actuator	roller slip, backlash, drive compliance, valve friction, latency	●●● high	measure delivered motion; backlash compensation [84]; learned transmission model [65]*	low once measured; high until measured (H5: measure first)	engineering	P IC-001*, IC-007*
Device mechanics	bending and torsional stiffness, tip shape, device-to-device variation	●●○ medium	tensile tests [52]*; per-device fit [12]; torsion rig [82]	medium; torsional lag partly intrinsic	engineering	P IC-005*, IC-004* C IC-002*
Contact and friction	wall and lumen friction, stick-slip [86], valve friction in Fz	●○○ low–medium	tuned friction ([13], [1]*); lumped friction from proximal Fz/Tz; formal fit: not found	high: history-dependent; Fz sums friction along the path	intrinsic (part)	C IC-002*, IC-008* P IC-007* *none: path-summed Fz*
Anatomy	geometry error, tortuosity, wall compliance, pulsatility, flow	●●● phantom ●○○ in vivo	phantom STL, rigid walls [1]*; many-anatomy training [2]*; motion randomisation [45]	low on the bench; medium–high in vivo	engineering	P IC-004* D IC-012*, IC-011*
Perception	X-ray noise; roll and depth (one plane); tracker not validated [5]	●●○ medium	low-dimensional state and history in the observation [3]*; simulated delay [13]	medium; roll and depth from one plane: intrinsic	intrinsic (part)	C IC-003*, IC-016* D IC-018* *none: roll and depth*
Workflow	contrast timing, view changes, device exchanges, takeovers	●○○ low	explicit state machine; human in the loop; not measured in any study read	low for task 1 (fixed device and view); high in vivo	engineering	C IC-013*

Dots: how far the component can be measured on the bench. Heavier outline: intrinsic (part), an observability or physical limit. P predicts (improves the simulator before use); C corrects online from feedback; D detects and stops or hands over. No study measured the components separately; the split is reasoned, not measured (D4).

b



c

